Stochastic Processes 2021/22 Semester 1

Tutorial 3

1. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the state space $S = \{0, 1, 2\}$ and has transition matrix

$$P = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 1 & 0 & 0 \\ 1/4 & 1/2 & 1/4 \end{pmatrix}.$$

   (a) i. Find the matrix giving the 2-step transition probabilities, and the matrix giving the 3-step transition probabilities.

   ii. If the chain is in state 1 at time $n = 5$, what is the probability that it is in state 0 at time $n = 7$?

   (b) Suppose that $\mathbf{P}(X_0 = i) = 1/3$ for $i = 0, 1, 2$.

   i. Find the distribution of the random variable X_3.

   ii. Find also

$$\mathbf{P}(X_0 = i \,|\, X_3 = 2), \qquad i = 0, 1, 2,$$

   i.e. find the initial distribution of the chain *given* that the chain is in state 2 after 3 steps.

2. A taxicab driver moves between the airport A and two hotels B and C according to the following rules. If he is at the airport, he will be at one of the two hotels next with equal probability. If at a hotel, then he returns to the airport with probability 3/4 and goes to the other hotel with probability 1/4.

   (a) Find the transition matrix for the Markov chain which corresponds to the driver's movements between A, B, and C.

   (b) Suppose that the driver starts the day at A. Determine the probability that after 3 journeys, the driver is at hotel C. If after 10 journeys the driver is back at the airport (A), what is the probability that the driver is at hotel C at the end of the 13th journey?

3. A Markov chain $\{X_n\}_{n\geq 0}$ takes values in the state space $S = \{1, 2\}$ and has transition matrix

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1 & 0 \end{pmatrix}.$$

   (a) Show by induction that the matrix of n-step transition probabilities is given by

$$P^n = \begin{pmatrix} 2/3 & 1/3 \\ 2/3 & 1/3 \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} 1/3 & -1/3 \\ -2/3 & 2/3 \end{pmatrix}.$$

   (b) Deduce that

$$\lim_{n\to\infty} p_{ij}^{(n)} = \begin{cases} 2/3 & \text{if } j = 1 \\ 1/3 & \text{if } j = 2, \end{cases}$$

   independently of the state i.

   (c) Suppose that $\mathbf{P}(X_0 = i) = 1/2$ for $i = 1, 2$.

   i. Find an expression for $\mathbf{P}(X_n = i)$ for $i = 1, 2$ and all n.

   ii. Find also $\lim_{n\to\infty} \mathbf{P}(X_n = i)$ for $i = 1, 2$.

   (d) Suppose that $\mathbf{P}(X_0 = 1) = 2/3$ and $\mathbf{P}(X_0 = 2) = 1/3$. Find $\mathbf{P}(X_n = i)$ for $i = 1, 2$ and all n.

4. In unprofitable times a corporation may sometimes suspend dividend payments. Suppose that a corporation usually pays a dividend 4 times a year (i.e. quarterly). Also, suppose that if the corporation pays a dividend in one quarter then it will pay a dividend in the next quarter with probability 0.9, whereas if the corporation doesn't pay a dividend in one quarter then the dividend will be suspended in the next quarter with probability 0.4.

   (a) Given that the corporation paid a dividend for the first quarter of 2003, what is the probability that the corporation will pay a dividend in the last quarter of 2003?

   (b) Again, given that the corporation paid a dividend for the first quarter of 2003, what is the probability that the corporation will pay a dividend in the last quarter of 2005?

5. For each of the Markov chains whose transition matrix is given below, determine the classes of intercommunicating states. Identify the closed classes and the vector of absorption probabilities associated with each of these. (In each case assume the states are labelled $1, 2, \ldots$.)

   (a)
$$\begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/6 & 1/2 \\ 0 & 0 & 3/4 & 1/4 \\ 0 & 1/2 & 1/2 & 0 \end{pmatrix}$$

   (b)
$$\begin{pmatrix} 0 & 1/2 & 1/2 & 0 \\ 1/3 & 0 & 1/6 & 1/2 \\ 0 & 0 & 3/4 & 1/4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

   (c)
$$\begin{pmatrix} 0 & 0 & 1/2 & 0 & 0 & 1/2 & 0 \\ 1/4 & 0 & 0 & 0 & 1/4 & 1/2 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1/6 & 0 & 0 & 1/6 & 1/6 & 0 & 1/2 \\ 1/2 & 0 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

6. A fair coin is tossed repeatedly with results $Y_0, Y_1, Y_2, \ldots$ that are 0 (for a tail) and 1 (for a head), with probability 1/2 for each.

   For $n \geq 1$ let $X_n = Y_n + Y_{n-1}$ be the number of heads in the $(n-1)$th and nth tosses. In other words, $X_n = 2$ if the $(n-1)$th and nth tosses were both heads, $X_n = 0$ if the $(n-1)$th and nth tosses were both tails etc.

   Is $\{X_n\}_{n \geq 1}$ a Markov chain?